

NATCHITOCHES, LA, USA

WMO: 722319

Lat: 31.736N

Lon: 93.099W

Elev: 37

StdP: 100.88

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	-1.3	-8.4	1.9	1.2	-5.9	2.3	3.1	9.3	10.5	8.4	10.8	1.2	0	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	37.5	26.3	36.2	26.1	35.0	25.9	28.6	33.7	27.7	32.7	27.2	32.1	2.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.5	23.5	30.2	26.4	21.9	29.5	26.0	21.5	29.3	92.2	33.8	88.2	33.4	85.5	31.9	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.1	5.8	DB	-5.8	38.9	2.4	2.1	-7.5	40.4	-8.9	41.6	-10.2	42.7	-11.9	44.2	
			WB	-6.5	28.5	2.3	1.1	-8.2	29.3	-9.5	29.9	-10.8	30.5	-12.5	31.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	9.2	11.1	15.5	19.0	23.9	27.3	28.1	28.7	25.7	20.0	14.0	10.4
	DBStd	8.11	5.39	5.49	5.00	4.00	3.69	2.16	1.85	2.07	2.66	4.53	5.05	5.16
	HDD10.0	219	82	47	12	1	0	0	0	0	0	1	18	58
	HDD18.3	1094	287	211	116	39	5	0	0	0	0	37	147	253
	CDD10.0	3669	57	78	182	271	432	518	562	579	472	311	138	71
	CDD18.3	1501	3	8	27	59	178	268	303	321	222	89	17	7
	CDH23.3	15451	10	42	204	530	1823	2823	3284	3594	2123	862	126	29
	CDH26.7	7415	0	3	32	123	828	1433	1684	1942	1022	327	19	2
Wind	WSAvg	2.2	2.6	2.8	2.7	2.8	2.3	1.8	1.5	1.5	1.6	1.8	2.2	2.5
Precipitation	PrecAvg	1347	117	119	118	107	148	108	88	87	86	100	108	152
	PrecMax	2076	348	286	223	359	465	403	199	184	194	302	372	553
	PrecMin	812	14	35	20	8	29	19	5	4	14	0	15	21
	PrecStd	314	81	69	57	85	101	84	55	48	46	76	74	95
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.9	27.1	29.3	31.5	37.6	38.9	38.0	39.0	36.7	33.8	28.9	26.3
		MCWB	19.1	20.0	21.1	22.9	25.4	26.5	26.5	26.5	24.4	23.2	21.8	21.5
	2%	DB	22.6	24.8	27.5	29.8	34.1	36.7	37.0	37.5	35.1	32.2	27.0	23.7
		MCWB	17.9	19.0	20.8	22.7	24.1	26.4	26.6	26.4	24.9	23.0	20.8	19.9
	5%	DB	21.0	22.4	26.0	27.8	32.6	35.0	35.2	36.2	33.8	30.4	24.8	22.1
		MCWB	17.2	18.5	19.6	20.8	24.1	26.3	26.3	26.3	24.8	22.1	19.6	19.3
	10%	DB	18.7	20.4	23.7	26.7	31.3	33.0	33.9	34.9	32.4	28.0	22.5	20.0
		MCWB	15.3	17.5	18.6	20.4	23.8	25.8	26.1	26.1	24.3	21.3	19.0	17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.4	21.8	23.5	24.8	27.2	29.4	29.2	29.8	28.2	26.0	23.1	22.7
		MCDB	23.1	24.5	27.0	28.2	30.8	32.3	32.6	34.1	31.6	29.3	26.2	25.0
	2%	WB	19.7	20.6	22.2	23.7	26.0	27.9	28.0	28.8	27.0	24.9	22.2	21.2
		MCDB	21.4	22.9	25.7	27.5	31.5	32.6	33.1	34.4	31.0	29.3	25.1	22.9
	5%	WB	17.9	19.2	21.0	22.9	25.3	27.3	27.4	27.8	26.1	23.8	21.1	20.0
		MCDB	19.8	21.8	23.9	26.3	30.9	32.4	32.5	33.1	30.7	28.2	23.5	21.7
	10%	WB	16.4	17.9	20.0	22.0	24.6	26.6	26.9	27.1	25.4	22.8	19.7	17.9
		MCDB	18.0	20.0	23.0	25.1	30.0	31.8	31.9	32.1	29.9	26.2	21.9	19.2
Mean Daily Temperature Range	5% DB	MDBR	11.7	11.4	12.6	12.8	12.5	11.5	11.4	12.2	12.3	13.6	13.0	11.4
		MCDBR	14.0	13.3	13.7	13.9	13.9	14.0	13.3	13.8	13.8	15.1	14.2	13.0
		MCWBR	9.3	8.4	7.5	6.9	5.3	5.0	4.3	4.3	4.9	6.5	8.8	9.2
	5% WB	MCDBR	11.3	11.3	11.5	11.5	11.8	12.2	11.6	12.4	11.4	11.6	11.4	11.2
	MCWBR	8.9	8.1	7.4	6.7	5.5	5.4	4.8	5.0	4.9	5.8	8.0	8.9	
Clear-Sky Solar Irradiance	taub	0.331	0.348	0.370	0.408	0.439	0.475	0.490	0.488	0.436	0.368	0.355	0.339	
	taud	2.518	2.471	2.408	2.307	2.269	2.208	2.198	2.200	2.316	2.480	2.468	2.494	
	Ebn at Noon	895	910	911	884	855	820	805	803	834	876	856	863	
	Edh at Noon	86	99	113	130	137	145	146	144	123	97	90	85	
All-Sky Solar Radiation	RadAvg	2.73	3.09	4.34	5.36	5.94	6.14	6.12	5.78	5.02	4.17	3.09	2.39	
	RadStd	0.26	0.53	0.36	0.39	0.41	0.46	0.47	0.50	0.47	0.58	0.32	0.29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	+69

Nomenclature: See separate page